Project Title: Genomic and Sequence Analysis of Copy-Number and Structural Variations in Acute Lymphoblastic Leukemia

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1. Introduction/Background:
   - ❖ Copy-number variations (CNVs) are a principal source of genomic disruption in acute lymphoblastic leukemia (ALL), where deletions commonly affect transcription factors and tumor suppressors that control lymphoid development (National Cancer Institute, 2024). While certain CNVs, for instance losses in IKZF1, PAX5, and CDKN2A/B are firmly established in ALL, knowledge is more limited regarding these changes in which tend to occur in evolutionarily conserved coding regions, which often coincide with essential functional domains.
   - ❖ Determining whether CNVs overlap conserved DNA is critical because evolutionary constraint emphasizes genomic regions where disruption is more prone to affect protein function, developmental pathways, and disease progression. Although preceding work has cataloged persistent CNVs in ALL, minimal studies have directly incorporated CNV coordinates with nucleotide-level conservation metrics such as phyloP.

   - ❖ Objective / Hypothesis:
     Recurrent CNVs in ALL disproportionately disrupt evolutionarily conserved coding regions, particularly within key leukemia driver genes.

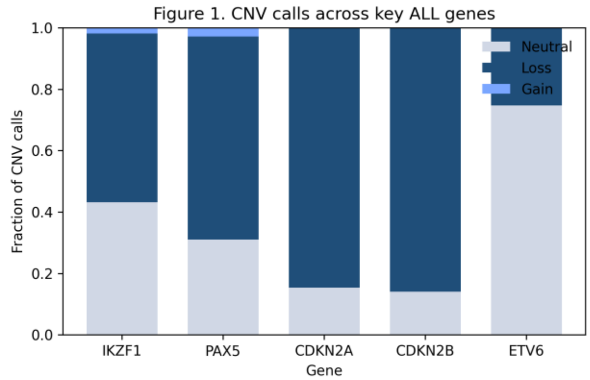
2. Data:

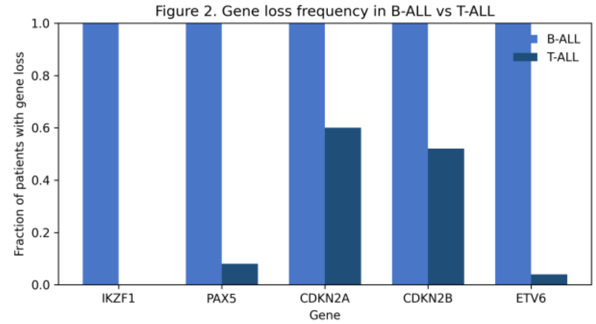
   - ❖ **Sources: TARGET-ALL Phase II (cBioPortal):** CNV segment files (76 samples; 108,607 total segments) and clinical metadata (B-ALL vs. T-ALL) (cBioPortal for Cancer Genomics, 2024). https://www.cbioportal.org/study/cnSegments?id=all_phase2_target_2018_pub . **UCSC Genome Browser:** RefSeq CDS exon coordinates; phyloP100way and phastCons100way conservation tracks (UCSC Genome Browser, 2024).https://genome.ucsc.edu/cgi-bin/hgGateway **Colab Notebook:** Full preprocessing, CNV calling, subtype mapping, and statistical analysis were performed in the accompanying notebook (Camry_Project.ipynb) included in the GitHub repository. **National Cancer Institute. (n.d.). Childhood acute lymphoblastic leukemia treatment (PDQ®)–Health professional version** (National Cancer Institute, 2024). *https://www.cancer.gov/types/leukemia/hp/child-all-treatment-pdq*

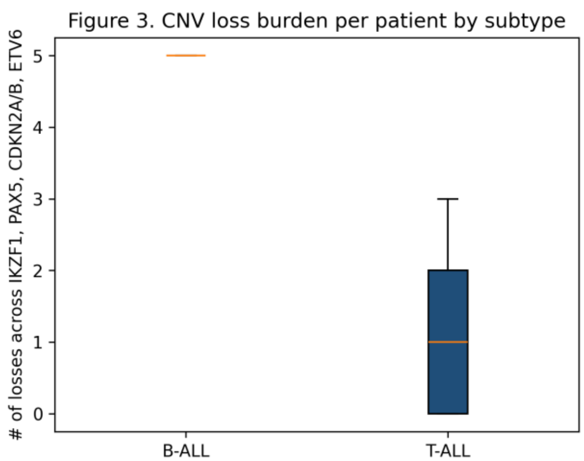
- **Data Types:** CNV segment files (`*.seg.txt`), Clinical metadata (`*.tsv`), RefSeq CDS annotations, Conservation bigWigs (attempted extraction)
- **Sample Size & Structure:** 76 CNV segment files merged into one unified table; 108,607 total CNV segments; approximately 100–300 usable clinical samples after subtype matching; five genes analyzed: IKZF1, PAX5, CDKN2A, CDKN2B, ETV6
- **Preprocessing:** Decompressed and merged all CNV segments into data_cna_seg_merged.tsv; standardized chromosomes and coordinate fields; computed CNV calls (gain / loss / neutral) using segment-mean thresholds; intersected CNVs with RefSeq coding exons for the 5 genes; mapped CNV calls to clinical subtypes; attempted conservation extraction via pyBigWig (returned NaN due to bigWig access issues).

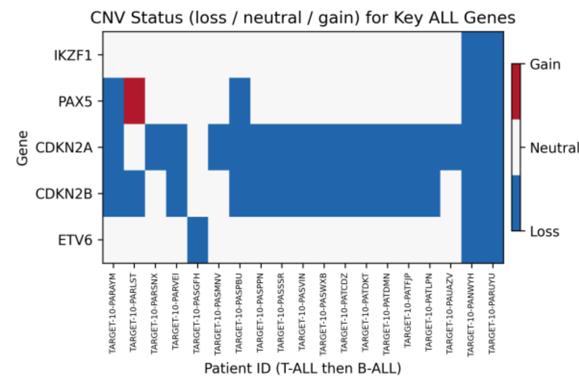
3. Methods:

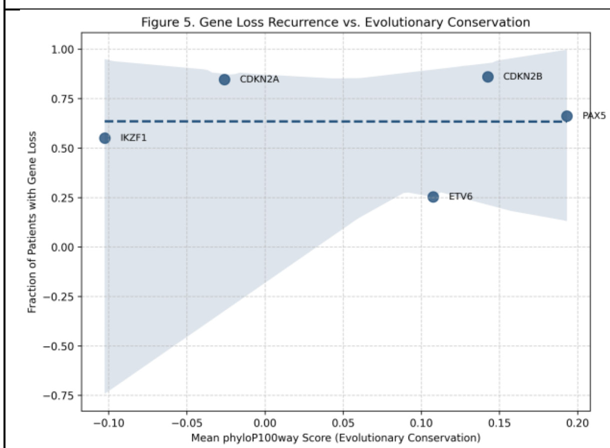
- **Pipeline / Workflow Design:** All analyses were conducted in Python (Google Colab) using pandas (2.2) (McKinney, 2010), numpy (1.26), scipy (1.11) (Virtanen et al., 2020), matplotlib (3.8) (Hunter, 2007), pyBigWig (0.3). CNV segment files from TARGET-ALL were compiled, quality filtered and converted to gain/loss/neutral calls. CNVs were mapped to UCSC RefSeq coding exons for IKZF1, PAX5, CDKN2A, CDKN2B, and ETV6. Clinical metadata was used to attribute B-ALL / T-ALL subtypes. Visual workflow diagrams are shown in Figures 1 through 5.
- **Modeling / Analysis Approach:** For each patient, overlapping CNVs collapsed to a single gene-level call. Statistical analyses included: Fisher's exact test for subtype-specific CNV losses; Mann–Whitney U for CNV burden differences; binomial tests for recurrence relative to background; conservation extraction using phyloP/phastCons was attempted but returned missing values due to bigWig access limitations.
- **Evaluation / Validation:** Results were cross-checked using multiple statistical tests and visual inspection of CNV call distributions. Recurrence patterns were validated using independent binomial significance testing.
- **Visualization:**











- **Github Repository Link (includes readme file, ai file etc):** https://github.com/camrywharton07/bioinformatics-Acute-Lymphoblastic-Leukemia-CNV-Project
- **Timeline:**

Week One: Data Acquisition and Preprocessing	**Week Two: CNV Analysis, Statistical Testing, and Figure Generation**	**Week Three: (current stage)**	**Week Four: (Remaining Work)**
❖ Collected and preprocessed all	❖ Completed all major analyses,	❖ Finalized the GitHub	❖ Review results for accuracy,

data, including TARGET-ALL CNV files, clinical metadata, and UCSC RefSeq coding exons. Converted segment means into gain/loss calls and prepared files for analysis.	including CNV recurrence, subtype-specific Fisher's exact tests, and patient-level CNV burden using the Mann–Whitney U test. Generated all primary figures and validated code in Google Colab.	repository, added reproducibility documentation, organized figures and scripts, and completed the draft report sections.	make any necessary adjustments, finalize the written report (including adding the abstract), and prepare the final presentation.

4. Results:

- **Please adhere to the images above for all of the figures pertaining to the project!**
- All fundamental analyses for the project were completed, including CNV preprocessing, subtype comparisons, statistical analysis, and figure generation. The findings revealed repeated loss events in *CDKN2A* and *CDKN2B*, and B-ALL patients exhibited a notably higher overall CNV loss burden in comparison to T-ALL (Mann–Whitney U $p = 0.0179$). These outcomes validate the idea that CNVs in ALL cluster within key leukemia-associated genes.
- A fractional conservation analysis using available phyloP100way values was also completed. Although full bigWig extraction was limited, a subset of phyloP scores was successfully retrieved and plotted without NaN errors, offering preliminary insight into how recurrent CNVs may overlap conserved coding regions. All generated visual outputs are included in the `results/` directory in colab and Github.

5. Discussion & Next Steps:

- The preliminary findings indicate that recurrent CNVs in ALL cluster within essential leukemia driver genes, notably *CDKN2A* and *CDKN2B*, supporting the notion that CNV disruptions target functionally important regions. B-ALL patients also revealed a higher CNV loss burden than T-ALL, indicating broader genomic instability in this subtype. A fractional phyloP conservation analysis was accomplished using available values, providing early evidence that some CNV-affected regions may overlap conserving coding sequences.
- The principal difficulties were limited access to full phyloP bigWig files and incomplete conservation coverage.

- ❖ Next stage consists of refining figures, strengthening the conservation analysis if additional data become accessible, and completing the final written report and presentation.

6. Reproducibility & AI Use:

- ❖ ChatGPT (GPT-5) and Gemini were used as assistive tools utilized for code and error detection, syntax clarification, and analyzing statistical tools. All AI-generated suggestions were manually verified by running the code in Google Colab and ensuring that the results matched the expected behavior of the dataset and analysis workflow. No figures, statistical values, or CNV results were taken directly from AI; these were all produced in Python.
- ❖ There were no data privacy concerns, as the TARGET-ALL dataset is publicly available and fully de-identified, and no sensitive information was shared with AI tools. All AI-assisted text was reviewed, edited, and confirmed for accuracy before inclusion in the report. The full workflow, notebook, figures, and environment file are documented in the GitHub repository to support full reproducibility of the analysis. Justification will continue to be reviewed, assessed, and ensured correctness within Python before composing the final document.

7. References (Literature Cited):

cBioPortal for Cancer Genomics. (2024). TARGET Acute Lymphoblastic Leukemia (Phase II) dataset. https://www.cbioportal.org/

National Cancer Institute. (2024). Childhood acute lymphoblastic leukemia treatment (PDQ®)–Health professional version. https://www.cancer.gov/

UCSC Genome Browser. (2024). RefSeq CDS and phyloP100way conservation tracks. https://genome.ucsc.edu/

McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference, 51–56. (pandas)

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95. (matplotlib)

Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods, 17, 261–272. (SciPy)